
Test Plan Document

Library management system

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GROUP LMS

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Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
ASP	Active Server Pages
TP	Test Plan
GUI	Graphic user interface
DFD	Data flow diagramm
QA	Quality assurance
OS	Operating system
FR	Functional requirements
UC	Use case
Integration Testing	Testing combined modules to ensure they work together properly.
Unit testing	Testing individual modules of the system separately to ensure correct functionality.
System testing	Testing the complete system to verify overall functionality and performance.
Traceability Matrix	A table that maps requirements to test cases to ensure full test coverage.

Table of Contents

1. Introduction.....	4
1.1 <i>Purpose of Document</i>	4
1.2 <i>Project Overview</i>	4
2. Scope of Testing.....	4
3. Test Plan Strategy.....	5
3.1 <i>Unit Testing</i>	5
3.2 <i>Integration Testing</i>	5
3.3 <i>System Testing</i>	5
4. Test Environment.....	5
5. Schedule.....	5
6. Control Activities.....	6
7. Functions to be Tested.....	6
8. Functions not to be Tested.....	7
9. Test Case Design and Description.....	7
10.Traceability Matrix.....	8
11.Major Deliverables.....	9
12.Risks and Assumptions.....	9
13.Exit Criteria.....	9
14.References.....	9
15.Appendices.....	10

1. Introduction

Library Management System is a software application developed to do all tasks related to a library including addition, deletion, issue, return, and organize books. In this test plan, it is described how testing work will be done in order to check if it is working.

1.1 Purpose of Document

The aim of this document is to describe the testing methodology, coverage, resource allocation, and testing schedule for library management system. It also ensures that all the functional and non functional requirements are tested.

It is targeted for developers, test team members, project managers, and supervisors.

1.2 Project Overview

- 1. Library Management System enables administrators to handle library tasks, its management.*
- 2. The system allows users (students, staff, and librarians) to easily search for books, view availability, and track issued or returned books through a user-friendly interface.*
- 3. It increases efficiency, saves time, and enhances communication, management and organization within library services.*

2. Scope of Testing

In Scope

- User login and authentication
- Book management (add, update, delete, issue, return and categorize books)
- Member management (students, staff, and other library users)
- Book issue and return management
- Tracking/searching book for availability and track issued records
- Notification management (for return book and overdue fine)
- Library schedule management (working hours, issue/return timings)
- Maintenance of library records and data (like maintaining books record)
- Report generation (issued books, returned books, overdue records)

Out of Scope

- Automatic fine calculation if book is returned late
- Online payments for overdue fines
- Online Book search, issue and reservation
- Book return Notification
- Fine notification if book is returned after date of return

3. Test Plan Strategy

"The aim of this test plan is to verify the Library Management System for all functional as well as non-functional requirements specified in the SRS,"

3.1 Unit Testing

Definition: It includes tests on individual modules and checks whether they work correctly like Login, Add book, Add user, issue book, return book and search books using different credentials like search by name, author name, edition, publisher name, published year.

Participants:

Zenab noor, Shahzeen fatima, Saba maryam

Methodology:

Each module is tested by the developers after it is developed
Errors are repaired before integration.

3.2 Integration Testing

Definition: The integration tests that the interactions. This includes verifying the interaction between login management, book management, user management, book issue/return, search book functionality, and notifications for overdue book return and fine notification management to make sure data flows correctly across the system.

Participants:

Zenab noor, Shahzeen fatima, Saba maryam

Methodology:

Modules integrated cumulatively
Dataflow across the system tested using Dataflow Diagram (DFD)

3.3 System Testing

Definition: System testing: System testing is used for verifying the entire library management system.

Participants:

Zenab noor, Shahzeen fatima, Saba maryam

Methodology: *End-to-end testing*

4. Test Environment

Hardware: PC/laptop

Barcode scanner to issue book

OS: Windows/Android

Software: Web Browser (Chrome)

Database: MySQL

Network: Internet Connection

5. Schedule

Testing Activities	Begin	End
Designing Test Cases	12-10-2025	17-10-2025
Executing Test Cases	20-10-2025	24-10-2025
Unit Testing	23-10-2025	27-10-2025
Integration Testing	28-10-2025	29-10-2025
System Testing	-----	-----

6. Control Activities

Regular review meetings

Attendees:DR.ILYAS

Error assessment and status update

7. Functions to be Tested

7.1 User Login:

Verification of user authentication using valid and invalid credentials.

7.2 Add / Update Book:

Testing the ability to add new books and update existing book details by adding different credentials of a book.

7.3 Add / Update User Information:

Check the validation of adding and modifying user records.

7.4 Issue Book:

Testing the process of issuing books to registered users.

7.5 Return Book:

Verification of correct book return processing and record updates.

7.6 Search Book:

Testing book search functionality using book title, author name, publisher, and published year.

7.7 Return Reminder Notifications:

Testing notification alerts sent to users as reminders to return books on time.

7.8 Fine Notifications:

Verification of notifications sent when a book is returned after the due date.Also check the automatic calculation of fine if book is returned after due date.

7.9 Schedule Management:

Testing library schedules such as issue/return timings and working hours.

7.10 Maintenance Management:

Verification of system and record maintenance features.

7.11 Report Generation:

Testing the generation of reports related to books, users, issued items, and overdue records.

8. Functions not to be Tested

8.1 Online Payment System:

Online payment functionality for fines or other services will not be tested.

8.2 Offline SMS Alerts:

Offline SMS-based notification services are not covered under testing.

8.3 Hardware internals:

Internal working and configuration of hardware devices (such as barcode scanners or other peripheral devices for the system) are not included in testing.

9. Test Case Design and Description

9.1 Test Case ID: TC-01

Objective: User Login

Step No	Execution Description	Procedure Result
1	Enter valid username and password	Credentials accepted
2	Click Login button	User dashboard displayed

9.2 Test Case ID: TC-02

Objective: Add New Book

Step No	Execution Description	Procedure Result
1	Enter book details	Data accepted
2	Click Save	Book record created

9.3 Test Case ID: TC-03

Objective: Issue Book

Step No	Execution Description	Procedure Result
1	Select available book	Book selected
2	Assign book to user	Book issued successfully

Test Case ID:	1	QA Test Engineer	Shahzeen fatima
Test case Version:	1.0	Reviewed By	Zenab noor ,saba maryam
Test Date:	Date	Use Case Reference(s)	Relation to use cases
Revision History	2 revisions done		
Objective	Verify that authorized users can successfully log in to the Library Management System.		
Product/Ver/Module	Library Management System (LMS) / Version 1.0 / Login Module		
Environment:	Windows 10/11, Web Browser, LMS Application, Database Server		
Assumptions:	User account exists and database connection is active.		
Pre-Requisite:	LMS application is running and user has valid login credentials.		
Step No.	Execution description	Procedure result	
1	1.Enter valid username and password System accepts credentials 2 Click Login button User dashboard is displayed	passed	
Comments			
Verify successful login and access to LMS features.			
		☐ Passed ☐ Failed ☐ Not Executed	

10. Traceability Matrix

Sr. No	Requirement ID	Use Case ID	GUI ID	Test Case ID	Status	Date Completed
1	FR-01	UC-01 Login	GUI-01	TC-01	Complete	20-10-2025
2	FR-02	UC-02 Add Book	GUI-02	TC-02	Complete	22-10-2025
3	FR-03	UC-03 Issue Book	GUI-03	TC-03	Complete	23-10-2025

11. Major Deliverables

- *Test plan*
- *Test cases*
- *Test summary report*
- Test Execution Report

12. Risks and Assumptions

Assumptions:

- Internet connection is available.
- Library database is accessible.
- Users have valid login credentials.

Risks:

- Database failure may affect transactions.
- Network issues may delay notifications.
- Incorrect data entry may impact records.

13. Exit Criteria

1. All critical test cases passed.
2. No high-priority defects remaining.
3. System meets all functional requirements.
4. Book issue and return functions work correctly.

14. References

Ref. No.	Document Title	Date of Release/ Publication	Document Source
BTS-SRS-01	Software Requirements Specifications	2025	Project Repository
LMS-SRS-01	Software Requirements Specification	2025	Project Repository
LMS-TP-01	Test Plan Document	2025	Project Repository

15. Appendices

15.1 Appendix A: Sample Test Cases

- Login Test Case
- Add Book Test Case
- Issue Book Test Case

15.2 Appendix B: Test Data

- Sample Student Records
- Sample Book Records
- Sample Admin Accounts